Assignment 9 - Doxygen

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Chapter 1

Assignment 9 - Doxygen Documentation

1.1 Introduction

This project demonstrates the use of Doxygen for generating documentation of a multi-file C++ project.

1.2 Features

- Logging messages with different severity levels.
- Mathematical calculations using CalculationEngine.
- Documentation of classes, functions and enumerations.

Chapter 2

Todo List

Member CalculationEngine::power (double base, int exponent)

Extend this function to support additional mathematical operations and optimized algorithms.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
: Performs mathematical calculations 9

Logger
: Provides logging functionality for the application 10

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

CalculationEngine.h
: Defines the CalculationEngine class 13

Logger.h
: Defines the Logger class for application logging 14

main.cpp
: Main entry point of the application 15

Chapter 5

Class Documentation

5.1 CalculationEngine Class Reference

Performs mathematical calculations.

`#include <CalculationEngine.h>`

Public Member Functions

- CalculationEngine (Logger &log)
  Constructs a CalculationEngine object.
- double power (double base, int exponent)
  Calculates the power of a number.

5.1.1 Detailed Description

Performs mathematical calculations.

The CalculationEngine class provides mathematical operations and uses the Logger class to display calculation information.

Currently, it supports power calculation using a base and an integer exponent.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            Logger & log)  [inline]
```

Constructs a CalculationEngine object.

Initializes the calculation engine with a Logger object.

Parameters

log	Reference to the Logger object.

5.1.3 Member Function Documentation

5.1.3.1 power()

```
double CalculationEngine::power (
            double base,
            int exponent)  [inline]
```

Calculates the power of a number.

Calculates base raised to the given exponent. Supports both positive and negative integer exponents.

Parameters

base	The base value.
exponent	The integer exponent.

Returns

The calculated power result.

Todo Extend this function to support additional mathematical operations and optimized algorithms.

Exceptions

invalid_argument	If base is zero and exponent is negative.

The documentation for this class was generated from the following file:

- CalculationEngine.h

5.2 Logger Class Reference

Provides logging functionality for the application.

```
#include <Logger.h>
```

Public Types

- enum class LogLevel { INFO , WARNING , ERROR }

  Defines the available logging levels.

Public Member Functions

- Logger (LogLevel level=LogLevel::INFO)

  Constructs a Logger object.
- void log (const string &message)

  Displays a log message.

5.2.1 Detailed Description

Provides logging functionality for the application.

The Logger class displays messages with different logging levels, including INFO, WARNING and ERROR.

5.2.2 Member Enumeration Documentation

5.2.2.1 LogLevel

```
enum class Logger::LogLevel  [strong]
```

Defines the available logging levels.

Enumerator

INFO	General informational messages.
WARNING	Warning messages.
ERROR	Error messages.

5.2.3 Constructor & Destructor Documentation

5.2.3.1 Logger()

```
Logger::Logger (
            LogLevel level = LogLevel::INFO)  [inline]
```

Constructs a Logger object.

Initializes the logger with the specified logging level.

Parameters

level	Initial logging level. Default is INFO.

5.2.4 Member Function Documentation

5.2.4.1 log()

```
void Logger::log (
            const string & message)  [inline]
```

Displays a log message.

Prints the message along with its logging level.

Parameters

message	The message to be displayed.

The documentation for this class was generated from the following file:

- Logger.h

Chapter 6

File Documentation

6.1 CalculationEngine.h File Reference

Defines the CalculationEngine class.

```
#include "Logger.h"
#include <stdexcept>
```

Classes

- class CalculationEngine

  Performs mathematical calculations.

6.1.1 Detailed Description

Defines the CalculationEngine class.

6.2 CalculationEngine.h

Go to the documentation of this file.

```
00001
00006
00007 #ifndef CALCULATION_ENGINE_H
00008 #define CALCULATION_ENGINE_H
00009
00010 #include "Logger.h"
00011 #include <stdexcept>
00012
00013 using namespace std;
00014
00026 class CalculationEngine
00027 {
00028 private:
00029
00030     Logger& logger;
00031
00032 public:
00033
00041     CalculationEngine(Logger& log)
00042         : logger(log)
```

```
00043     {
00044         logger.log("Calculation Engine initialized.");
00045     }
00046
00063     double power(double base, int exponent)
00064     {
00065         logger.log("Performing power calculation.");
00066
00067         if (base == 0.0 && exponent < 0)
00068         {
00069             throw invalid_argument(
00070                 "Zero cannot have a negative exponent."
00071             );
00072         }
00073
00074         double result = 1.0;
00075         long long exp = exponent;
00076
00077         if (exp < 0)
00078         {
00079             exp = -exp;
00080         }
00081
00082         for (long long i = 0; i < exp; i++)
00083         {
00084             result *= base;
00085         }
00086
00087         if (exponent < 0)
00088         {
00089             result = 1.0 / result;
00090         }
00091
00092         logger.log("Power calculation completed.");
00093
00094         return result;
00095     }
00096 };
00097
00098 #endif // CALCULATION_ENGINE_H
```

6.3 Logger.h File Reference

Defines the Logger class for application logging.

```
#include <iostream>
#include <string>
```

Classes

- class Logger

  Provides logging functionality for the application.

6.3.1 Detailed Description

Defines the Logger class for application logging.

6.4 Logger.h

Go to the documentation of this file.

```
00001
00017
00018 #ifndef LOGGER_H
00019 #define LOGGER_H
00020
00021 #include <iostream>
00022 #include <string>
00023
00024 using namespace std;
00025
00033 class Logger
00034 {
00035 public:
00036
00041     enum class LogLevel
00042     {
00043         INFO,
00044         WARNING,
00045         ERROR
00046     };
00047
00055     Logger(LogLevel level = LogLevel::INFO)
00056         : currentLevel(level)
00057     {
00058     }
00059
00067     void log(const string& message)
00068     {
00069         cout « "[" « getLevelName() « "] "
00070              « message « endl;
00071     }
00072
00073 private:
00074
00075     LogLevel currentLevel;
00076
00082     string getLevelName() const
00083     {
00084         switch (currentLevel)
00085         {
00086             case LogLevel::INFO:
00087                 return "INFO";
00088
00089             case LogLevel::WARNING:
00090                 return "WARNING";
00091
00092             case LogLevel::ERROR:
00093                 return "ERROR";
00094         }
00095
00096         return "UNKNOWN";
00097     }
00098 };
00099
00100 #endif // LOGGER_H
```

6.5 main.cpp File Reference

Main entry point of the application.

```
#include "Logger.h"
#include "CalculationEngine.h"
#include <iostream>
```

Functions

- int main ()

  Main function of the application.

6.5.1 Detailed Description

Main entry point of the application.

This file demonstrates the usage of Logger and CalculationEngine classes.

6.5.2 Function Documentation

6.5.2.1 main()

```
int main ()
```

Main function of the application.

Creates a Logger and CalculationEngine object, performs a power calculation and displays the result.

Returns

0 if the program executes successfully.